

Knowledge-Based Agent (KBA) in Artificial Intelligence

What is a Knowledge-Based Agent?

A **Knowledge-Based Agent** is an intelligent system that:

- Uses **knowledge** about the world
- Applies **logic**
- Makes **decisions**
- Takes **actions**

Example:

A robot knows “If it’s raining, take an umbrella” → So it picks up an umbrella before going outside.

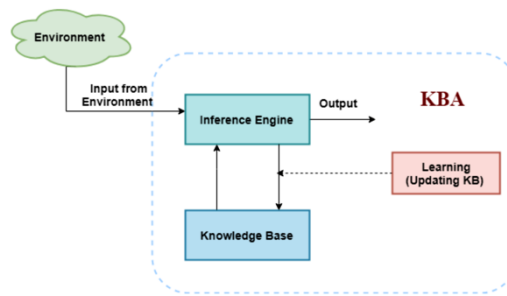
Key Features of KBA:

-  **Stores knowledge**
 - Example: “If fever → give medicine”
 -  **Applies logic to reason**
 - Example: “Temperature > 100°F → fever → give paracetamol”
 -  **Updates knowledge from new perceptions**
 - Example: “Road blocked → find alternate route”
 -  **Understands internal state of the world**
 - Example: “Door open at night → security risk”
 -  **Chooses appropriate actions**
 - Example: “Close the door”
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Main Components of KBA:

Component	Role	Example
 Knowledge Base (KB)	Stores facts and rules	“If traffic → reroute”
 Inference System	Applies logic to KB	“Traffic detected → take another road”

The architecture of knowledge-based agent:



KBA-এর ভিতরের গঠন (Architecture)

- **Perception** → Agent পরিবেশ থেকে তথ্য নেয় ➤ যেমন: "রাস্তায় গর্ত"
- **Inference Engine** → KB থেকে সিদ্ধান্ত নেয় ➤ যেমন: "গর্ত আছে → গতি কমাও"
- **Learning Element** → নতুন তথ্য শিখে KB আপডেট করে ➤ যেমন: "নতুন রাস্তা পাওয়া গেছে → KB-তে যোগ করো"

What is Knowledge Base?

- A collection of **sentences** (facts/rules)
 - Written in **Knowledge Representation Language**
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- • KB হলো agent-এর **মস্তিষ্কের মতো**, যেখানে সে তার জানা সব তথ্য রাখে।
 - • এখানে তথ্য থাকে **sentence** আকারে, যেমন: ➤ "IF traffic THEN reroute"

What is Inference System?

Inference System হলো একটি বুদ্ধিমান সিস্টেমের অংশ, যা নিয়ম (rules) ও তথ্য (facts) ব্যবহার করে নতুন সিদ্ধান্ত বা উপসংহার তৈরি করে। এটি যুক্তি প্রয়োগ করে নতুন জ্ঞান তৈরি করে।

- Derives **new knowledge** from existing facts
- Uses **logical rules** to reason

 Example:

"IF dirt detected
THEN clean area"

Two types of reasoning:

-  **Forward Chaining** → From facts to conclusion
➤ "Rain → take umbrella → go outside"
-  **Backward Chaining** → From goal to required facts
➤ "Want to go outside → need umbrella → is it raining?"



Operations of KBA:

1. **TELL** → Inform KB about perception
 - “Patient has fever”
 2. **ASK** → Ask KB what to do
 - “What medicine for fever?”
 3. **PERFORM** → Do the action
 - “Give paracetamol”
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Generic KBA Program (Simplified):

```
function KB-AGENT(percept):  
    TELL(KB, percept)  
    Action = ASK(KB, action)  
    TELL(KB, action_done)  
    return action
```



Real-Life Examples:

- **Smart Fan Agent**

- percept: room temperature = 32°C
- action: turn on fan

- **Cooking Robot**

- percept: rice = uncooked
- action: boil rice

- **Security Agent**

- percept: door = open at night
 - action: close door
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Levels of a Knowledge-Based Agent

A Knowledge-Based Agent can be understood at **three distinct levels**:

1. Knowledge Level – *What the agent knows and what its goals are*

At this level, we define:

- The agent's knowledge about the world
- The goals it wants to achieve

Example:

A medical robot knows that fever means high body temperature and that paracetamol reduces fever.

→ Knowledge: "Fever = temperature > 100°F"

→ Goal: "Make the patient healthy"

2. Logical Level – *How the knowledge is represented logically*

At this level, the agent's knowledge is encoded using logical rules or sentences.

Example:

The medical robot represents its knowledge as:

- IF body_temperature > 100°F THEN fever
- IF fever THEN give(paracetamol)

→ The agent uses these logical rules to reason and make decisions.

3. Implementation Level – *How the agent acts in the real world*

At this level, the agent physically performs actions based on its knowledge and reasoning.

Example:

The medical robot measures the patient's temperature, finds it to be 102°F, and gives paracetamol.

→ The agent applies its logic and executes the action in reality.

Approaches to Designing a Knowledge-Based Agent

1. Declarative Approach

- Start with an empty knowledge base
- Add facts and rules gradually

Example:

Teach: “IF fever THEN give paracetamol”

2. Procedural Approach

- Directly write code for agent behavior

Example:

Program: If fever is detected, automatically give medicine

Knowledge Representation in AI

- It helps machines **think, understand, and act** using real-world knowledge.
 - It's more than storing data — it lets machines **learn and reason** like humans.
-

What AI Needs to Represent:

- **Objects** – Things and their features (e.g., guitar has strings)
 - **Events** – Actions that happen (e.g., concert occurred)
 - **Performance** – How to do tasks (e.g., cooking rice)
 - **Meta-knowledge** – Knowing what you know
 - **Facts** – True statements about the world
 - **Knowledge Base (KB)** – Collection of logical rules
 - **Knowledge** – Experience and awareness
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Types of Knowledge in AI (with Easy Examples)

1. **Declarative Knowledge**
 - Knowing *what* something is
 - Example: “A dog has four legs”
2. **Procedural Knowledge**
 - Knowing *how* to do something

- Example: “How to ride a bicycle”
 - 3. **Meta-Knowledge**
 - Knowing *about your own knowledge*
 - Example: “I know that I can cook”
 - 4. **Heuristic Knowledge**
 - Rules from experience (may not always be perfect)
 - Example: “If the computer is slow, restart it”
 - 5. **Structural Knowledge**
 - Knowing how things are connected
 - Example: “A wheel is part of a car”
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Relation between Knowledge and Intelligence

- **Knowledge is essential for intelligent behavior.**
- Without knowledge, an agent (or human) **cannot understand or act correctly.**
- AI agents use knowledge to **sense, reason, and make decisions.**
- Just like humans need experience to respond, AI needs knowledge to behave intelligently.

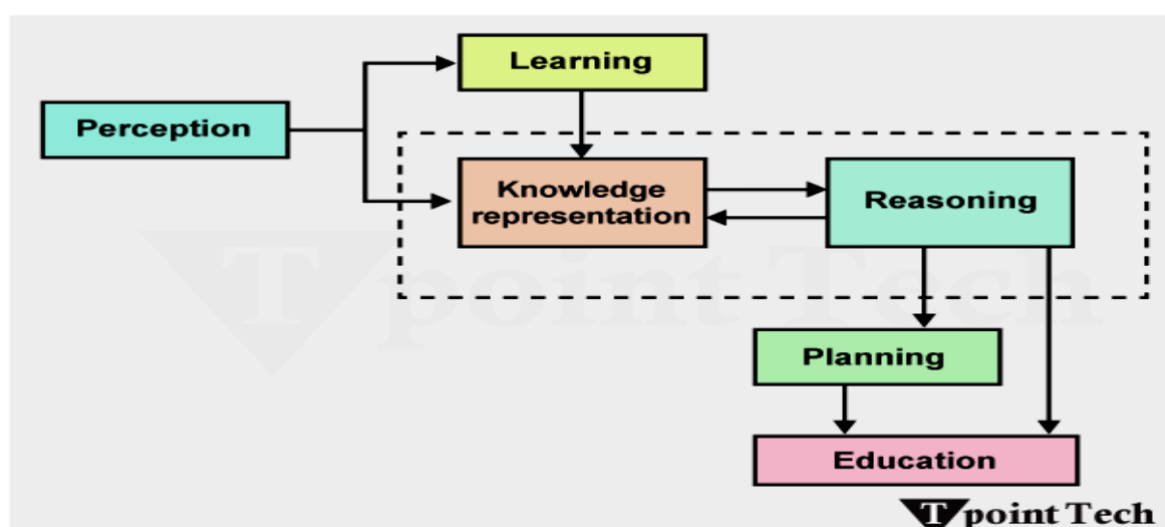
Example:

If a robot hears someone speaking in French but doesn't know the language, it can't respond.

→ Without knowledge, it can't act intelligently.

→ With knowledge of French, it can understand and reply.

AI Knowledge Cycle – Main Theme



An AI system shows intelligent behavior through a cycle of five key components:

1. Perception

- Collects data from the environment (e.g., images, sounds)

2. Learning

- Learns patterns and knowledge from the data

3. Knowledge Representation and Reasoning

- Stores knowledge and applies logic to make decisions
- These two are central to intelligent behavior

4. Planning

- Decides what actions to take based on reasoning

5. Execution

- Performs the chosen actions in the real world

Approaches to Knowledge Representation

AI systems use different methods to store and use knowledge. There are four main approaches:

1. Simple Relational Knowledge

- Stores facts in table format (rows and columns)
- Mostly used in databases
- **Example:**

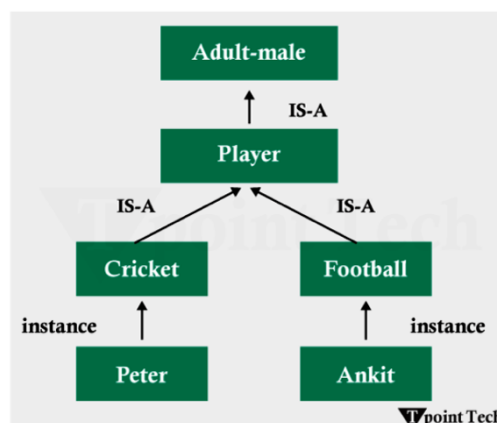
Player Weight Age

Player1 65 23

Player2 58 18

2. Inheritable Knowledge

- Uses a hierarchy of classes
- Objects inherit properties from parent classes
- **Example:**



3. Inferential Knowledge

- Uses logic to find new facts
- Helps AI reason and make conclusions
- **Example:**
 - Marcus is a man
 - All men are mortal
 - So, Marcus is mortal

4. Procedural Knowledge

- Describes how to do something using rules or code
- Uses “If-Then” logic
- **Example:**
IF water is boiling THEN add tea leaves

Requirements for a knowledge Representation system – Example: *Crossing the Road*

1. **Representational Accuracy**
 - Ability to store all necessary knowledge correctly
 - Example:* AI knows traffic rules like red light means stop
2. **Inferential Adequacy**
 - Ability to create new knowledge from existing facts
 - Example:* AI figures out “If light is red → don’t cross the road”
3. **Inferential Efficiency**
 - Ability to reason quickly and effectively
 - Example:* AI quickly decides to wait for green light
4. **Acquisitional Efficiency**
 - Ability to learn new knowledge easily
 - Example:* AI learns that some roads have countdown timers

Challenges in Knowledge Representation

1. **Ambiguity**

One word can have more than one meaning.

Example: “Bank” can mean a place for money or the side of a river.
2. **Uncertainty**

Sometimes information is not clear or complete.

Example: Predicting the stock market is hard because many things affect it.
3. **Scalability**

AI needs to handle a lot of data quickly and correctly.

Example: In healthcare or self-driving cars, there is a huge amount of data.
4. **Expressiveness vs Efficiency**

Too many details make AI slow. Less detail makes it fast but may miss things.

Solution: Use a mix of clear rules and learning from examples.

5. Dynamic Updates and Maintenance

Information changes often (like weather). AI needs regular updates.

Solution: Add new data step by step and fix wrong or repeated data.

Applications of Knowledge Representation

1. Solving Problems and Making Decisions

AI uses knowledge to break big problems into small parts and find smart solutions.

Example: Helping doctors find diseases, helping businesses make good decisions.

2. Robots and Automation

Robots use knowledge to understand their surroundings and do tasks.

Example: A robot can clean a room or drive a car by itself.

3. Healthcare and Industry

AI helps with medical advice, finds new medicines, and predicts when machines might break.

Example: Hospital software checks reports and suggests treatments.

4. Search Engines and Suggestions

AI helps platforms like Google or YouTube show better results and suggest things you like.

Example: If you search “Leonardo da Vinci,” it shows websites and also his life story.

Techniques of Knowledge Representation

1. Rule-Based System

→ Uses IF...THEN rules to make decisions

Example: If temperature > 100°C → Then water is boiling

2. Semantic Network

→ Shows relationships between concepts using connected links

Example: Cat → is a → Animal

Animal → has → Legs

3. Frame Representation

→ Stores knowledge in structured templates like forms

Example:

Frame: Car

- Brand: Toyota
- Color: Red
- Speed: 120 km/h

4. Ontology

→ Organizes knowledge into categories and defines how they relate

Example:

Fruit → Apple is a type of Fruit

Fruit → has color, taste, size

5. Logic-Based Representation

→ Uses logic to represent facts and draw conclusions

Example:

All humans are mortal

Socrates is a human

→ So, Socrates is mortal

What is Knowledge Representation?

- It helps machines understand and use human knowledge
 - Used in AI for reasoning, learning, decision-making, and understanding language
 - Makes intelligent systems smarter and more useful
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Types of Knowledge in Artificial Intelligence

AI systems need to understand and use different kinds of knowledge to think, learn, and make decisions. These types are:

1. Objects

- Refers to physical things in the real world
- Can be described by features like shape, color, size, etc.

Examples: Car, building, person

Used in: Object-oriented programming, image recognition

2. Events

- Refers to actions or activities that happen over time
- Helps AI understand sequences and changes

Examples: Driving a car, cooking a meal, attending a concert

Used in: Event-based systems, activity tracking

3. Performance

- Describes how well a system or agent completes a task
- Includes goals, methods, and evaluation criteria

Examples: Speed of a robot, accuracy of a prediction

Used in: Task planning, productivity analysis

4. Facts

- Statements that are either true or false
- Helps AI make logical decisions and draw conclusions

Examples: "Water boils at 100°C", "Earth orbits the Sun"

Used in: Knowledge-based systems, rule engines

5. Meta-Knowledge

- Knowledge about how other knowledge is organized and used
- Helps AI evaluate and manage its own knowledge

Examples: Knowing which source is more reliable, or how facts are linked

Used in: Reasoning systems, expert systems

6. Knowledge Base

- A structured collection of information that AI can access and use
- Acts like a digital brain for the system

Examples: Medical database, customer support FAQ

Used in: Chatbots, decision support systems

Knowledge Representation Techniques

AI uses different methods to store and understand human knowledge. The four main techniques are:

1. Logical Representation

→ Uses logic to express knowledge, often in “if...then...” format

Types of Logic:

- **Propositional Logic** – Deals with complete statements
- **Predicate Logic** – Deals with objects and their properties/relations

Example:

“All humans are mortal” → written as a logical rule

Advantages:

- Clear and accurate reasoning
- Used in many programming languages

Disadvantages:

- Can be complex to use
- Not easy for humans to read like natural language

2. Semantic Network Representation

→ Shows relationships between concepts using nodes and links

Example:

“Cat” → is a → “Animal”

“Animal” → has → “Legs”

Advantages:

- Easy to visualize connections
- Good for understanding meaning

3. Frame Representation

→ Uses templates (frames) to store structured knowledge

Example:

Frame: Car

- Brand: Toyota
- Color: Red
- Speed: 120 km/h

Advantages:

- Organizes data clearly
- Easy to update and reuse

4. Production Rules

→ Uses condition-action rules to make decisions

Format:

IF condition → THEN action

Example:

If temperature > 100°C → Then water is boiling

Advantages:

- Simple and flexible
 - Good for expert systems
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1. Semantic Network Representation

- Represents knowledge as a **graph**
- **Node** = object/concept, **Arc** = relationship

Common Relations:

- **IS-A** → Jerry IS-A Cat
- **Kind-of** → Cat Kind-of Mammal

Example Graph:

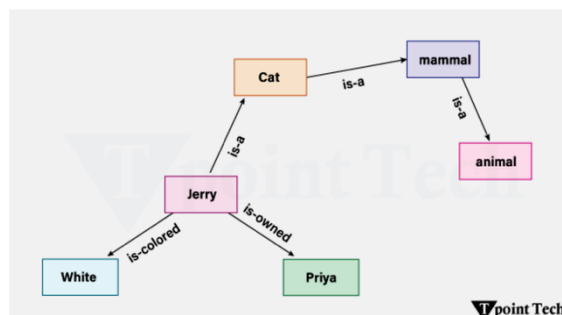
- Nodes: Jerry, Cat, Mammal, Priya, Brown, Animal
- Arcs: IS-A, owned-by, colour-of

✓ Advantages:

- Easy to understand
- Similar to human memory
- Easy to expand

✗ Disadvantages:

- Slow search
- No standard link naming
- Can't use "all" or "some"
- Not intelligent on its own — fully depends on the creator.



2. Frame Representation

- Stores knowledge in **slots** (attributes) and **facets** (extra info)
- Like a form or record

Example Frame – Book:

- Title: Artificial Intelligence
- Author: Peter Norvig
- Year: 1996
- Pages: 1152

Example Frame – Person:

- Name: Peter
- Profession: Doctor
- Age: 25

✅ Advantages:

- Easy to update and reuse
- Handles missing/default values
- Good for visualization

❌ Disadvantages:

- Complex reasoning
 - Not ideal for large systems
 - Too general sometimes
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3. Production Rules

- Uses **IF condition → THEN action** format
- Basis of rule-based systems

Main Parts:

1. Rules (IF-THEN)
2. Working Memory
3. Recognize-Act Cycle

Example Rules:

- IF (at bus stop AND bus arrives) → THEN get into bus
- IF (on bus AND unpaid) → THEN pay charges

 Advantages:

- Easy to add/change rules
- Can use natural language

 Disadvantages:

- No learning ability
 - Too many rules = slow system
-